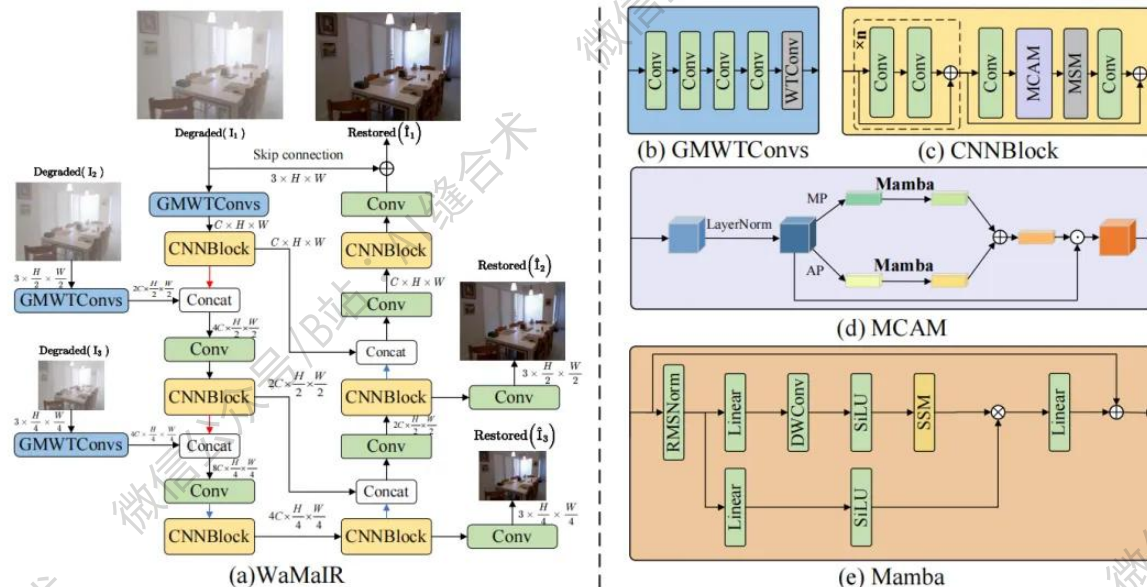


核心速览



论文题目: WaMaIR: Image Restoration via Multiscale Wavelet Convolutions and Mamba-based Channel Modeling with Texture Enhancement

中文题目: 基于纹理增强的多尺度小波卷积与 Mamba 通道建模的图像恢复方法

论文链接: <https://arxiv.org/pdf/2510.16765>

所属单位: 哈尔滨工程大学

核心速览: 本文提出了一种名为 WaMaIR 的新型图像修复框架, 通过多尺度小波卷积和基于 Mamba 的通道建模结合纹理增强技术, 显著提升了图像修复中纹理细节的重建质量与计算效率。

论文内容详细解读:

<https://mp.weixin.qq.com/s/dXNYuD8QhF-dPfdRKEjR0A>

```
GMWTConvs(  
  (conv_layers): Sequential(  
    (0): Conv2d(32, 16, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))  
    (1): BatchNorm2d(16, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
    (2): ReLU(inplace=True)  
    (3): Conv2d(16, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))  
    (4): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
    (5): ReLU(inplace=True)  
    (6): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))  
    (7): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
    (8): ReLU(inplace=True)  
    (9): Conv2d(32, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))  
    (10): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
    (11): ReLU(inplace=True)  
  )  
  (wtconv): WTConv(  
    (wt): HaarWaveletTransform()  
    (conv): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=4)  
    (iwt): InverseHaarWaveletTransform()  
  )  
)  
Input shape: torch.Size([1, 32, 256, 256])  
Output shape: torch.Size([1, 64, 256, 256])
```

哔哩哔哩/微信公众号: AI缝合术, 独家整理!